

QIZ Model: Progress Report

Preliminary

Research Question and Setting

Research question

Does duty-free US access under the Qualifying Industrial Zone (QIZ) programme raise Egyptian textile-firm performance, and can compliance with its rules of origin generate productivity upgrading that also improves performance in *non-US* markets?

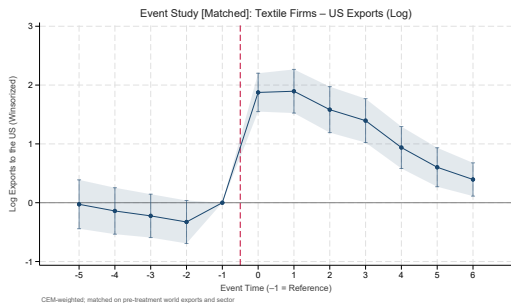
What is a QIZ?

- A bilateral trade arrangement between Egypt, Israel, and the United States (operative 2005).
- Egyptian firms in designated zones export to the US **duty-free** if at least $\gamma_s\%$ of input value comes from Israel.
- The mechanism is a package: tariff removal + a mandatory sourcing constraint (rules of origin, ROO).
- QIZs are concentrated in the textile and apparel sector; compliance requires firm-level certification.

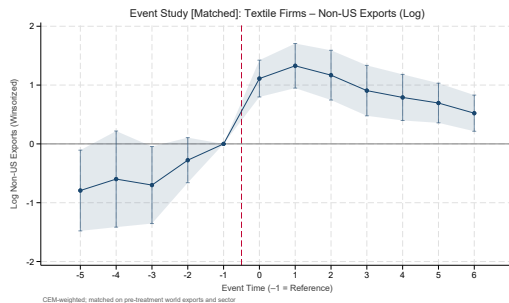
Why does the non-US margin matter?

A pure tariff story predicts gains only toward the US. Observing gains to non-US destinations as well suggests the ROO compliance process raises firm capability—not just market access.

Empirical Motivation



US exports rise sharply after QIZ-linked treatment.



Non-US exports also rise—motivating an upgrading / spillover channel.

These two facts must be reconciled in one framework.

Geography and sectors

- Two Egyptian regions: QIZ locations Q and non-QIZ locations N .
- Two sectors: textiles/apparel T and other manufacturing O .
- Three export destinations: Egypt (EG), United States (US), rest of world (RW).

Agents

- A representative household supplies labor and consumes a nested CES basket of varieties.
- A continuum of potential entrants in each region-sector; productivity is Pareto-distributed.
- Free entry drives expected profits to the entry cost f_{rs}^E .

Small open economy. Egypt takes US and RW demand and prices as given. Egypt cannot affect international prices. Domestic goods and labor markets clear; supply equals demand for the domestic variety and for labor.

Model: Household Side

Preferences. Household r maximises utility over a nested CES consumption bundle:

$$U_r = \left[\sum_s \beta_s^{\frac{1}{\rho}} C_{rs}^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}}, \quad C_{rs} = \left[\int_0^{M_{rs}} c(\omega)^{\frac{\sigma_s-1}{\sigma_s}} d\omega \right]^{\frac{\sigma_s}{\sigma_s-1}}$$

where β_s is the expenditure share on sector s , $\sigma_s > 1$ is the within-sector elasticity of substitution, and M_{rs} is the endogenous mass of available varieties.

Labor market. Households supply labor inelastically. Wages w_r are determined in equilibrium; labor is mobile across both sectors *and* regions, so a single wage w clears in equilibrium.

Price index.

$$P_{rs} = \left[\int_0^{M_{rs}} p(\omega)^{1-\sigma_s} d\omega \right]^{\frac{1}{1-\sigma_s}}$$

Real income and welfare are $W_r = w_r L_r / P_r^{\text{agg}}$.

Model: Production Side

Technology. A firm with productivity ϕ in region r , sector s uses labor ℓ and an intermediate bundle x (Cobb-Douglas):

$$q = \phi \ell^{\alpha_s} x^{1-\alpha_s}, \quad \alpha_s \in (0, 1).$$

Marginal cost is $c_{rs}(\phi) = \frac{1}{\phi} \left(\frac{w_r}{\alpha_s} \right)^{\alpha_s} \left(\frac{P_{rs}^x}{1-\alpha_s} \right)^{1-\alpha_s}$.

Market access. To sell to destination j , a firm pays a fixed cost f_{rjs} and an iceberg trade cost $d_{rjs} \geq 1$. Profits from market j :

$$\pi_{rjs}(\phi) = \frac{1}{\sigma_s} \left(\frac{c_{rs}(\phi) d_{rjs}}{P_{js}} \right)^{1-\sigma_s} E_{js} - f_{rjs}.$$

Entry. Potential entrants pay f_{rs}^E to draw $\phi \sim \text{Pareto}(\theta_s)$. The firm enters if $\Pi_{rs}(\phi) \equiv \max_{m \in \mathcal{M}_r} \left[\sum_j \pi_{rjs}^m(\phi) - F_{rs}^m \right] \geq 0$; otherwise it exits. Free entry pins down the equilibrium mass M_{rs} .

Model: The QIZ Mechanism

Who can comply? Only firms in QIZ regions ($r = Q$), textile sector ($s = T$), selling to the US.

Tariff removal. Non-compliers pay the US MFN tariff t_T^{MFN} . Compliers remove it, paying instead an Israeli-input sourcing cost. The effective price at the US border is:

$$p^{\text{US}}(\phi) = \begin{cases} c_{QT}(\phi) d_{QT}^{\text{US}} (1 + t_T^{\text{MFN}}) & \text{non-complier} \\ c_{QT}^{\text{ROO}}(\phi) d_{QT}^{\text{US}} & \text{complier} \end{cases}$$

where c^{ROO} reflects the Israeli-content requirement γ_T applied only to US-bound units (shipment-level certification).

ROO cost normalisation. The sourcing composite is normalised so that ROO does not impose a mechanical penalty beyond the required content share—it is only binding relative to cheaper domestic inputs.

Compliance margin. Firms differ in compliance fixed cost f_T^C drawn from a distribution, generating interior take-up. In equilibrium, a textile-QIZ firm complies iff $\pi_{QT,US}^{\text{complier}}(\phi) - f_T^C \geq \pi_{QT,US}^{\text{non-comp}}(\phi)$.

Upgrading. Compliers can pay a further cost to raise ϕ ; QIZ access raises the return to upgrading, generating gains that carry over to non-US destinations.

Calibration

Parameter	Method	Source / target moment
Sector shares β_s	IO absorption shares	Egypt IO table 2008
Labor shares α_s	Cost-share from IO	Egypt IO table 2008
Elasticity σ_s	Fixed from literature	Imbs-Méjean values
Pareto shape θ_s	Hill / log-rank on exporter sizes	Egyptian customs data
MFN tariffs t_s^{MFN}	Export-weighted averages	WTO tariff schedule
Trade wedges d_{rjs}	EK-style inversion	2004 trade shares
Labor endowments L_r	Governorate aggregation	Labor force survey 2004
Fixed costs $f_{\text{dom}}, f_{\text{US}}, f_{\text{RW}}$	Match export participation	2004 firm participation rates
Entry cost f_{rs}^E	Match Q/N firm-count ratios	Industrial statistics 2004
Compliance cost f_T^C	Match compliance rate	Customs uptake target: 32.1%
Upgrade block	Match non-US expansion	Event-study moments (customs)

Current fit (upgrade benchmark): model uptake **34.2%** vs. target **32.1%**; welfare gain **+0.23%**; QIZ raises upgrade intensity and generates non-US export gains.

Current Problems

1. Q-textile sector collapses in GE

Free entry drives Q-textile firm mass toward zero when calibrated to conditional exporter moments. The model fits compliance *among* US exporters while losing the Q-textile firm population altogether.

2. Aggregate US export margin is too weak

Matching complier-level US gains is consistent with aggregate all-sector US exports falling $\approx 2.7\%$ —the reallocation margin is wrong in sign.

3. Q location and US access are conflated

A Q-specific US trade-cost shifter does too much work: it is hard to separately identify the location advantage from the QIZ tariff channel.

Root cause

The model matches *conditional* moments well (uptake among US exporters) but is not disciplined on *unconditional* moments (firm mass, aggregate trade flows).

What I Want to Work On

Model fixes

- Separate the Q-location advantage from US tariff access so the two margins are identified independently.
- Add a discipline on Q-textile firm mass—either an incumbent-persistence device or an unconditional mass moment in the calibration.
- Re-calibrate f_{rs}^E jointly with unconditional firm-count moments, not only conditional exporter shares.

Empirical target

The next version must jointly match:

1. Treated textile firms expand to the US.
2. The same firms also expand to non-US destinations.
3. QIZ-region textile activity remains economically non-trivial in equilibrium.

Robustness

All results must be stable across productivity grid resolutions before reporting.

The Model in Equilibrium: A Verbal Summary

Firm decisions (a sequence of fixed costs)

1. **Entry.** Pay f_{rs}^E , draw productivity ϕ . Exit if expected profits are negative.
2. **Selling domestically.** Always the baseline option; no additional fixed cost.
3. **Exporting.** Pay a destination-specific fixed cost f_{US} or f_{RW} to access each foreign market. Only high- ϕ firms find it worthwhile.
4. **QIZ compliance.** A QIZ-region textile firm exporting to the US faces a choice: pay f_T^C to comply with the Israeli-content requirement and remove the MFN tariff, or stay non-compliant and pay the tariff. This is the central tradeoff of the model.
5. **Upgrading.** Pay a further fixed cost to raise productivity. QIZ access raises the return to upgrading—gains spill over to non-US destinations.

The core QIZ tradeoff

Compliance removes the US tariff—a variable cost saving that grows with export volume—but requires sourcing a share γ_T of inputs from Israel, raising marginal cost relative to cheaper domestic inputs. High-volume exporters to the US find compliance worthwhile; smaller exporters do not.

Equilibrium conditions

- **Free entry** in each region-sector pins down the endogenous firm mass.
- **Labor market clearing** across regions and sectors determines the equilibrium wage (labor is fully mobile).
- **Domestic goods market clears.** US and RW demand and prices are exogenous—Egypt is a small open economy and cannot affect